# K-FIELD
# CELL3 TO SPECTROPHOTOMETER
# NEAR-LOCK FACE ANGLE

Scale-independent dual-face metric correspondence between the measured pic_id14E plane pair and the harmonic Cell3 construction

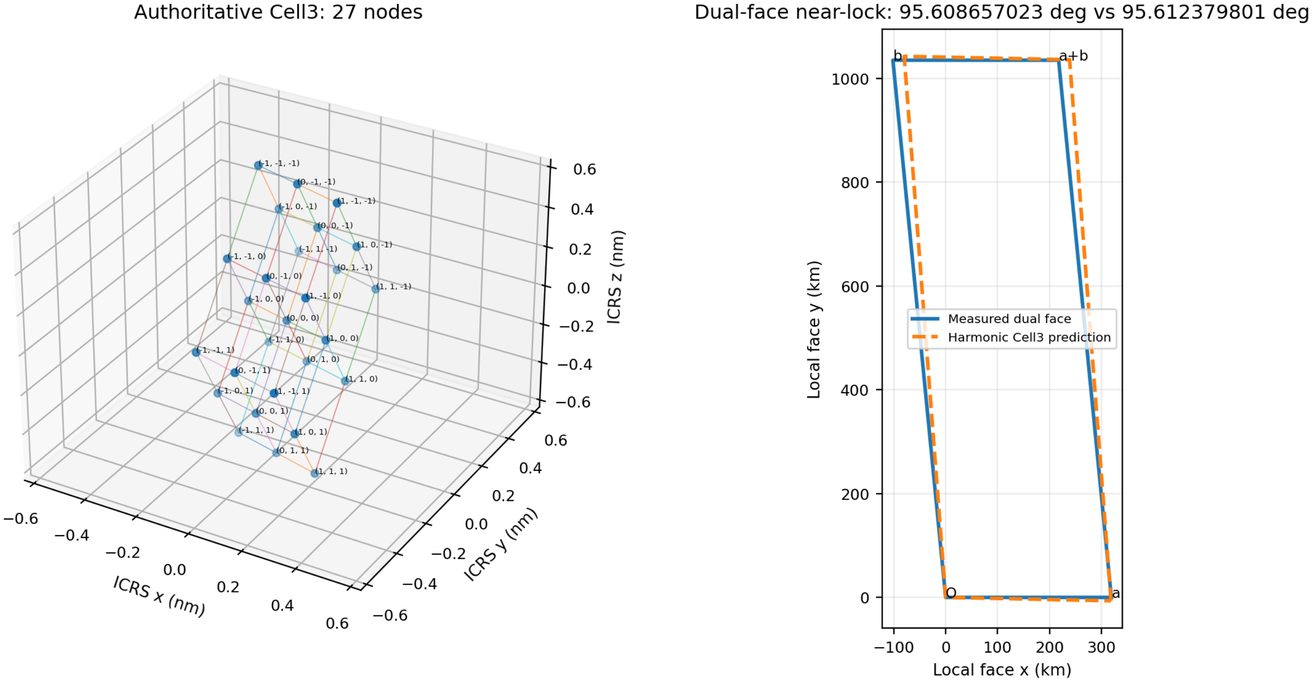


Figure 1. Data-derived title composite. Left: the authoritative Cell3 primitive geometry with all 27 centered half-step nodes. Right: the dimensionally recovered spectrophotometer dual face overlaid with its harmonic Cell3 prediction. Every coordinate is generated from the exact source ledgers.

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Archival publication epoch: 1785939329 | Generated UTC: 2026-08-05T14:15:29Z
Machine-readable companion: K-Field_cell3_to_spectrophotometer_near_lock_face_angle_1785939329.json
Rendered with Python ReportLab using Helvetica, Helvetica-Bold, Times-Roman, and Courier only.

# Abstract

This archival reference develops the exact geometric comparison between the dimensionally recovered spectrophotometer dual face and the harmonic Cell3 prediction. The measured included angle is 95.608657023 deg. The harmonic Cell3 included angle is 95.612379801 deg. Their difference is -0.003722778 deg, equal to -13.402001 arcseconds and -6.497473219029e-05 radians. The absolute separation is 10.341050 parts per million of a complete revolution.

The face angle is the principal comparison because it is dimensionless and invariant under any common direct-space scale factor. Edge lengths and area depend on recursive level and harmonic order. The internal angle depends only on the relative geometry of the two reciprocal plane families. The near-lock therefore tests the shape of the recovered dual metric rather than merely its selected scale.

The surrounding face metric is also reproduced: the measured direct edges are 319.393331 km and 1039.778275 km, compared with 318.962296 km and 1046.423190 km from the harmonic construction. The measured edge ratio is 3.255478981; the prediction is 3.280711244. The measured area is 3.305083742e+11 m^2; the prediction is 3.321695494e+11 m^2. These surrounding sub-percent agreements support the angle lock but do not replace it.

**Central result. The independently recovered spectrophotometer plane pair generates a direct-space dual face whose included angle is reproduced by the harmonic Cell3 prediction to 13.402000546 arcseconds. This scale-independent internal-metric lock is the principal geometric bridge between the measured macroscopic plane structure and the authoritative Cell3 geometry.**

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# 1. The Near-Lock as the Primary Comparable

The recovered plane structure contains several useful comparisons: individual plane orientations, plane spacings, spacing ratios, zone relations, reciprocal-vector sums, and the direct-space dual face. Among these, the complete dual-face angle is the most discriminating scale-independent quantity. It combines both measured plane families into one internal metric invariant.

```
Measured included angle = 95.608657023062 deg
Harmonic Cell3 prediction = 95.612379800991 deg
Measured minus prediction = -0.003722777929 deg
Angular residual = -13.402000546 arcsec
Angular residual = -6.497473219029302e-05 rad
```

The residual is 10.341049804 ppm of one complete 360-deg revolution and 38.936149662 ppm of the predicted included angle. Reversing either direct edge converts the obtuse angles to the equivalent acute pair 84.391342977 deg and 84.387620199 deg without changing the residual.

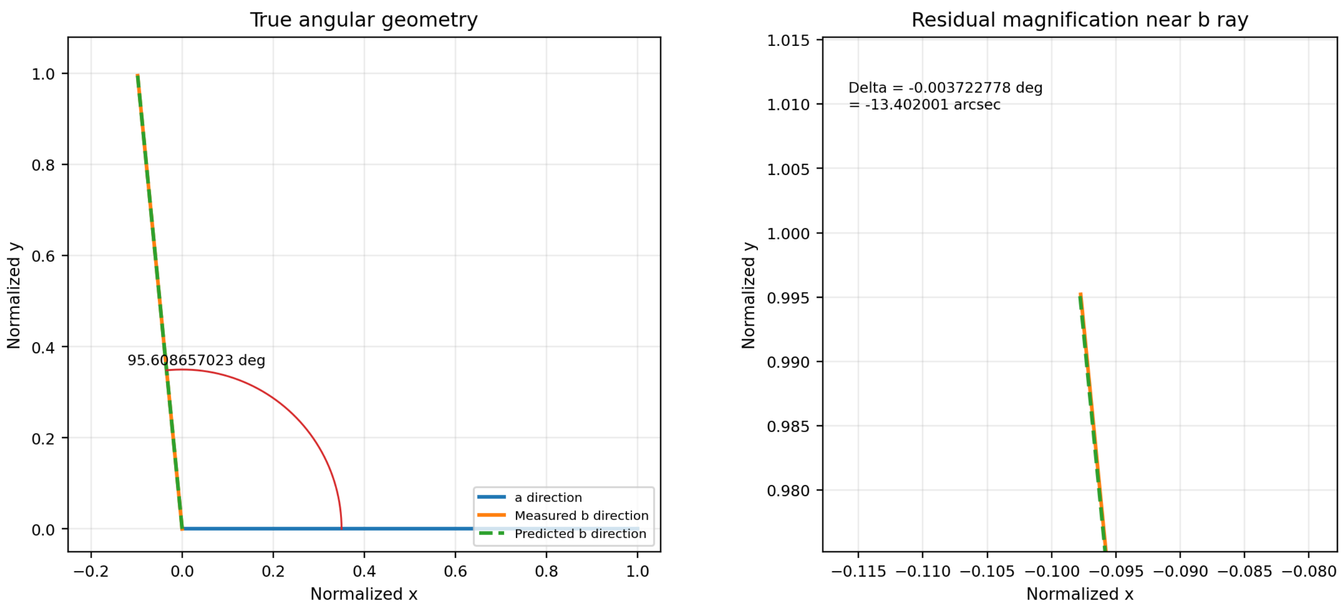


Figure 2. Exact dual-face angular comparison. Left: the measured and harmonic Cell3 rays in their true normalized geometry. Right: a local magnification around the b-edge direction showing the 13.402-arcsecond separation. The magnification changes only the view, not the calculated angle.

## 1.1 Why the full face is more important than either normal alone

The measured a edge differs from its harmonic prediction by 1.976316 deg, and the measured b edge differs by 2.873059 deg. The complete face normals differ by 2.928783 deg. Yet the angle between the two edges differs by only 0.003722778 deg. The individual orientation residuals are therefore correlated: the two edges move together in a way that preserves their internal separation.

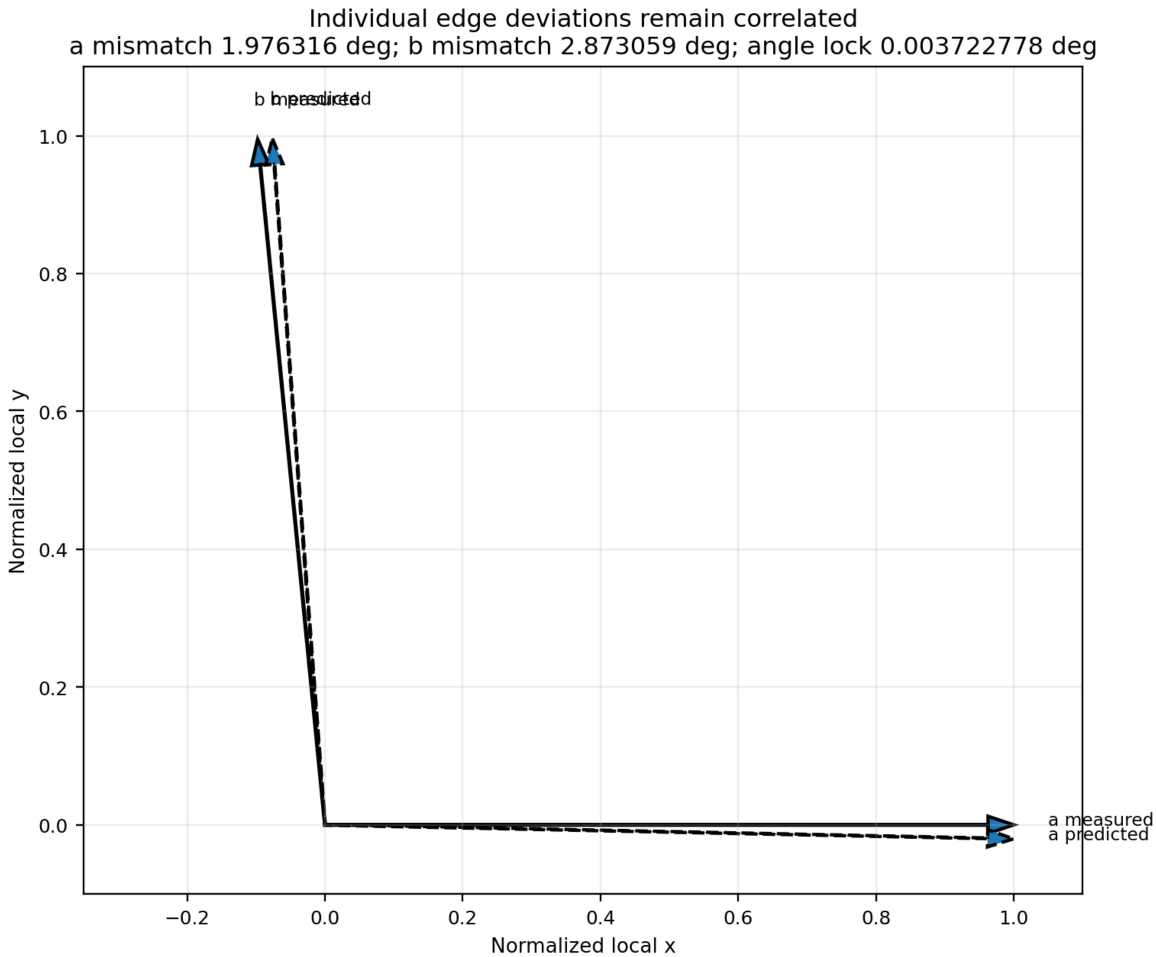


Figure 3. Correlated edge deviations. Each measured direct edge is compared with its harmonic Cell3 prediction after normalization. The individual vectors are not identical, but their relative angular separation is preserved to 0.003722778 deg.

**Interpretive statement. A pair of independently drifting directions would generally alter the internal face angle. The observed preservation identifies the pair as one coherent geometric unit in the recovered dual metric.**

# 2. Direct Space, Reciprocal Space, and the Dual Face

A direct-space vector translates between physical lattice points. A reciprocal vector labels parallel phase planes. Their units are inverse: metres for direct vectors and cycles per metre for reciprocal vectors. The dual-face construction converts two measured reciprocal plane families into the direct translations between their successive intersection nodes.

```
Q = [ qA^T ; qB^T ]
D = Q^T (Q Q^T)^(-1)
a_exp = first column of D
b_exp = second column of D
```

The construction satisfies four exact duality equations: qA.a_exp = 1, qB.a_exp = 0, qA.b_exp = 0, and qB.b_exp = 1. Moving by a_exp advances one qA cycle while preserving qB phase. Moving by b_exp advances one qB cycle while preserving qA phase.

```
a_exp = [23580.891690751218, 241209.358896143938, 208024.244963551173] m
b_exp = [320888.448477832426, -735846.439950803644, 660832.415677104844] m
```

The included angle is computed directly from the dot product:

```
theta = arccos( (a . b) / (||a|| ||b||) )
```

For the measured face, cos(theta) = -0.097733271091. For the harmonic Cell3 face, cos(theta) = -0.097797934560. The cosine residual is +6.466346870064e-05.

## 2.1 Scale invariance

If both direct edges are multiplied by any common scale s, the numerator and denominator of the cosine formula both acquire a factor s^2, which cancels. The angle is unchanged. Cell3 recursion changes direct lengths by 9^r, and harmonic order changes reciprocal spacing, but a common scaling does not change the internal face angle. This is why the near-lock is a cleaner structural comparison than absolute edge lengths.

```
angle(s a, s b) = angle(a, b) for every nonzero scale s
```

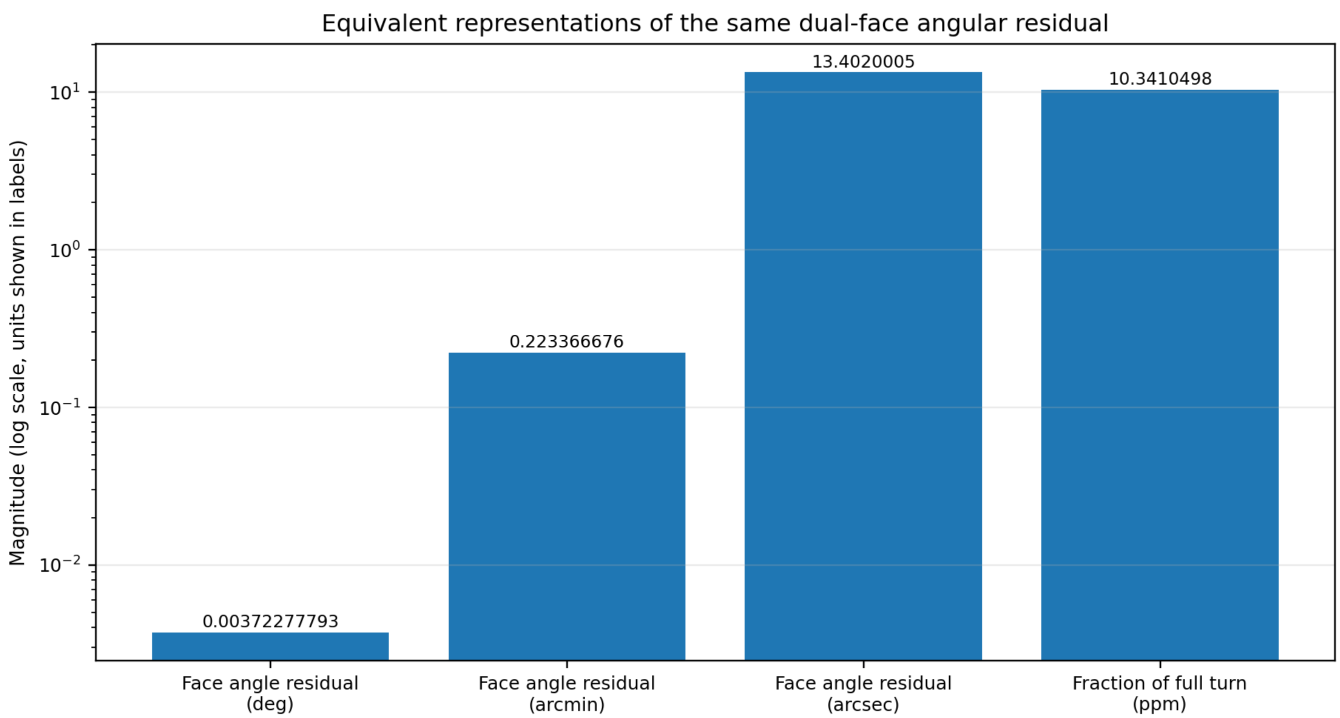


Figure 4. Equivalent numerical representations of the same angular residual. The degree, arcminute, arcsecond, and fraction-of-revolution values are not separate tests; they are different units for the single measured-predicted angle difference.

# 3. Authoritative Cell3 Direct Geometry

The authoritative Cell3 basis is fixed in ICRS Cartesian coordinates and nanometres. Its columns are the ordered direct vectors a, b, and c. No additional fitted rotation is applied.

```
B_nm = [[ 0.327501242056, 0.035330775844, -0.010129317627 ],
[ -0.054897007884, 0.207319850440, -0.389787702366 ],
[ -0.037487502895,-0.547401085389, -0.689734408367 ]] nm
```

| Vector | ICRS x (nm) | ICRS y (nm) | ICRS z (nm) | Length (nm) | RA (deg) | Dec (deg) |
|---|---|---|---|---|---|---|
| a | 0.327501242056 | -0.054897007884 | -0.037487502895 | 0.334179679059 | 350.484328 | -6.440865 |
| b | 0.035330775844 | 0.207319850440 | -0.547401085389 | 0.586410890412 | 80.328749 | -68.983472 |
| c | -0.010129317627 | -0.389787702366 | -0.689734408367 | 0.792319765042 | 268.511404 | -60.519741 |

| Primitive metric | Exact value |
|---|---|
| angle(a,b) | 83.933483012249 deg |
| angle(a,c) | 80.448130018322 deg |
| angle(b,c) | 50.363231813718 deg |
| A_ab | 0.194869165461 nm^2 |
| A_ac | 0.261106235790 nm^2 |
| A_bc | 0.357809539128 nm^2 |

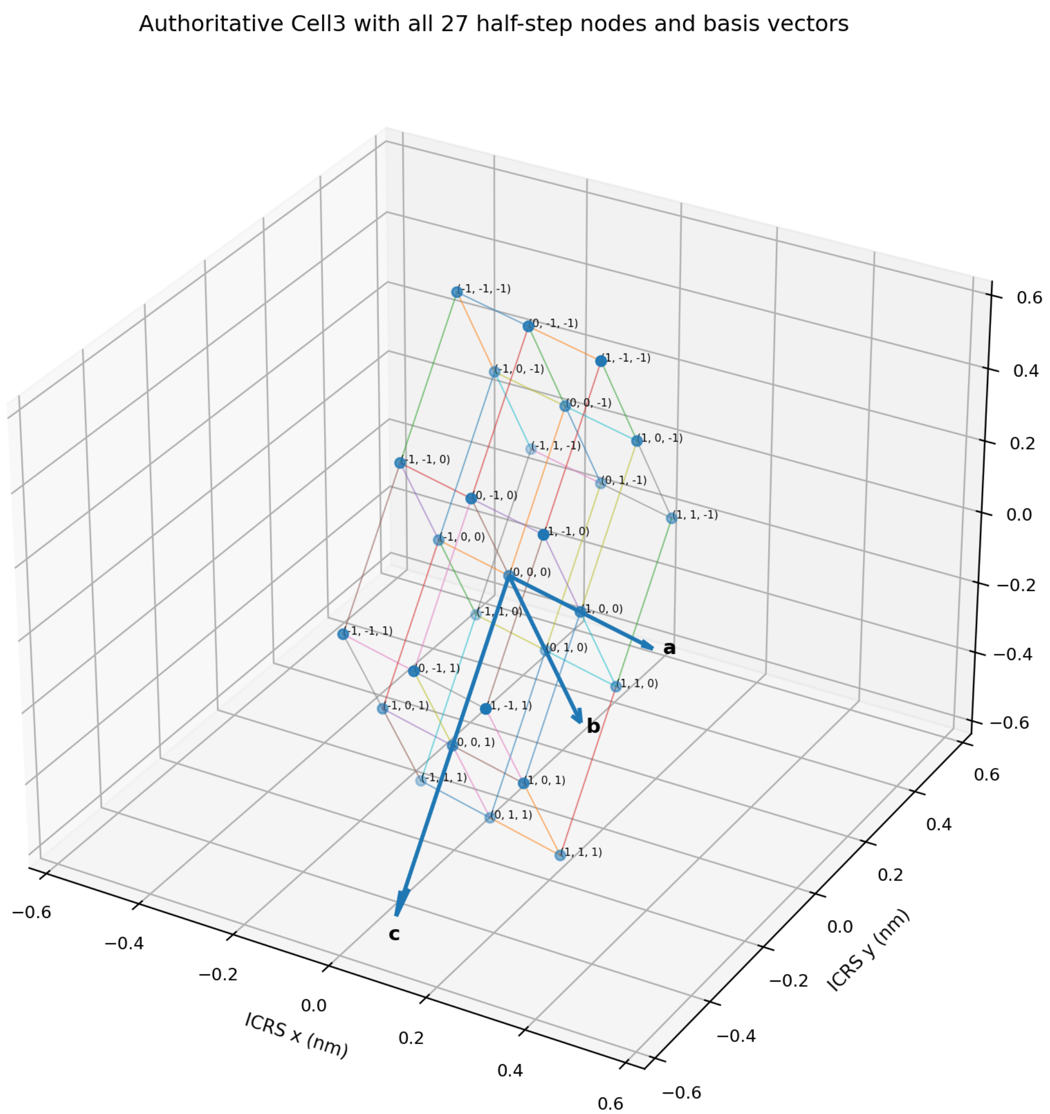


Figure 5. Authoritative Cell3 primitive volume with all 27 centered half-step nodes. Each address (i,j,k) represents direct fractional coordinate (i,j,k)/2. Nearest-neighbor links and the exact basis vectors a, b, and c are shown in ICRS nanometres.

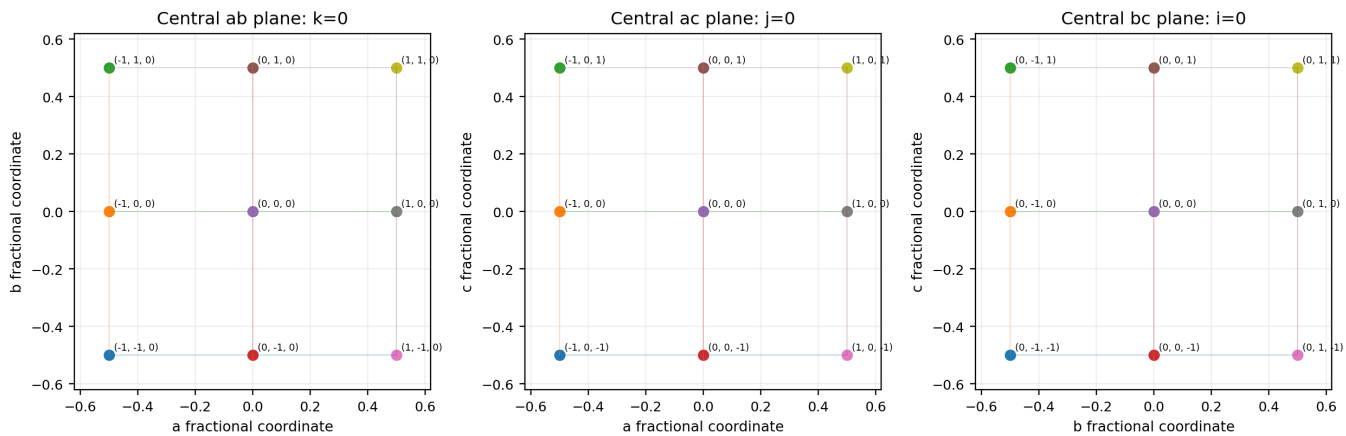


Figure 6. All nine nodes on each central Cell3 coordinate plane: ab, ac, and bc. These are dimensionless fractional-coordinate projections of the exact authoritative nodes, not schematic replacements.

The primitive Cell3 ab angle is 83.933483012 deg. The exposed dual-face acute angle is 84.391342977 deg, a difference of 0.457859965 deg. That primitive-angle proximity is a useful metric echo, but it is not the primary lock. The 13.402-arcsecond lock is with the specific high-index harmonic Cell3 face predicted from the recovered plane assignments.

# 4. Authoritative Reciprocal Geometry and Measured Planes

The reciprocal basis is G = B^(-T). For integer Miller packet (h,k,l), the reciprocal plane vector is G(h,k,l)^T and the primitive plane spacing is its inverse magnitude. Recursive Cell3 scaling changes the direct basis according to B_r = 9^r B. Harmonic order m multiplies the reciprocal vector.

```
g_hkl = B^(-T) [h k l]^T
q_hkl^(r,m) = (m / 9^r) g_hkl x 10^9 cycles/m
d_hkl^(r,m) = 1 / ||q_hkl^(r,m)||
```

| Mode | Spacing (km) | RA (deg) | Dec (deg) | Angle to CMB axis (deg) |
|---|---|---|---|---|
| qA | 317.864281 | 81.380625 | +45.795586 | 89.984183 |
| qB | 1034.800487 | 296.484362 | +44.633516 | 89.965843 |
| qC | 260.276750 | 46.645302 | +69.405455 | 89.996828 |
| qF | 670.885616 | 79.653162 | +48.332024 | 89.983619 |

The measured planes are not the primitive low-index Cell3 coordinate planes. They occupy high-index oblique reciprocal directions. The harmonic prediction used for the dual face assigns qA to the common family (1,6,19) at recursive level r=18 and harmonic order m=19, while the combination-optimized qB branch uses (1,-4,-2), r=18, m=18.

## 4.1 Exact measured and predicted reciprocal vectors

```
qA_measured = [3.287321093305e-07, 2.168679499253e-06, 2.255229490922e-06] cycles/m
qB_measured = [3.066748724203e-07, -6.155152953415e-07, 6.789419322445e-07] cycles/m

qA_Cell3 = [-2.391320425654e-07, -2.204364889077e-06, -2.237807284679e-06] cycles/m
qB_Cell3 = [3.447793754409e-07, -5.889564297487e-07, 6.755125843830e-07] cycles/m
```

Applying the same dual transformation D = Q^T(QQ^T)^(-1) to the measured pair and the harmonic Cell3 pair produces the two faces compared throughout this paper.

# 5. Dimensionally Recovered Dual Face with All Nodes

| Quantity | Measured face | Harmonic Cell3 face | Measured - prediction |
|---|---|---|---|
| a length | 319.393331 km | 318.962296 km | +0.135137% |
| b length | 1039.778275 km | 1046.423190 km | -0.635012% |
| included angle | 95.608657023 deg | 95.612379801 deg | -0.003722778 deg |
| acute equivalent | 84.391342977 deg | 84.387620199 deg | +0.003722778 deg |
| edge ratio b/a | 3.255478981 | 3.280711244 | -0.769110% |
| face area | 3.305083742e+11 m^2 | 3.321695494e+11 m^2 | -0.500099% |

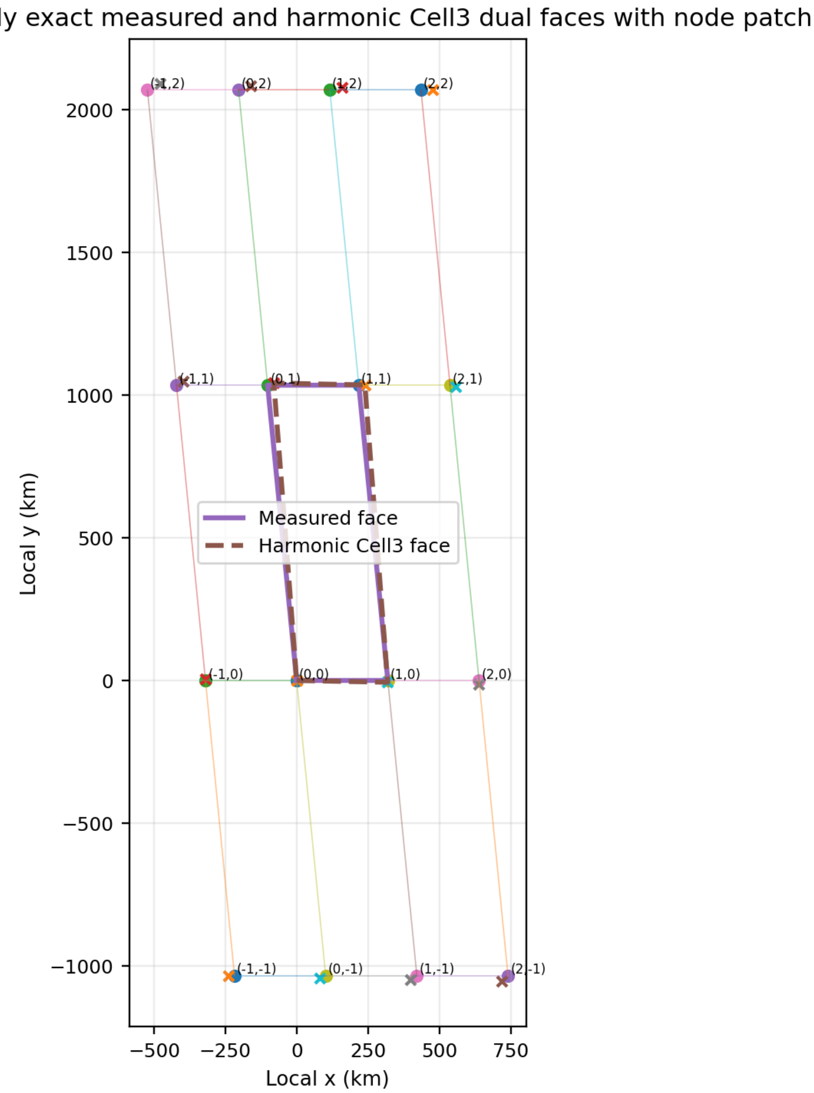


Figure 7. Measured and harmonic Cell3 dual faces with the complete local integer node patch -1 <= i,j <= 2. A node (i,j) is the intersection reached after i qA plane intervals and j qB plane intervals. Solid lines show measured geometry; dashed lines and x markers show the harmonic Cell3 prediction. Axes are kilometres.

The displayed base cell has four vertices O, a, b, and a+b. The surrounding 16-node patch demonstrates that the angle lock is not a local drawing convention. It propagates across every integer translation generated by the two direct edges.

## 5.1 Exact node phase property

```
r(i,j) = i a_exp + j b_exp
qA . r(i,j) = i cycles
qB . r(i,j) = j cycles
```

# 6. The Angle Lock in the Dual Metric Tensor

A two-dimensional face is completely characterized by its Gram tensor, the matrix of edge dot products. Normalizing the a edge to unit length isolates shape from overall scale. With r = |b|/|a| and included angle theta, the normalized tensor is

```
G_face = [[1, r cos(theta)],
[r cos(theta), r^2]]
```

The measured off-diagonal angular coupling is r cos(theta) = -0.318168610. The harmonic Cell3 value is -0.320846784. Their difference is +0.002678174. The normalized tensor relative Frobenius residual is 0.015247921; most of that residual comes from the 0.769-percent edge-ratio difference rather than the angle.

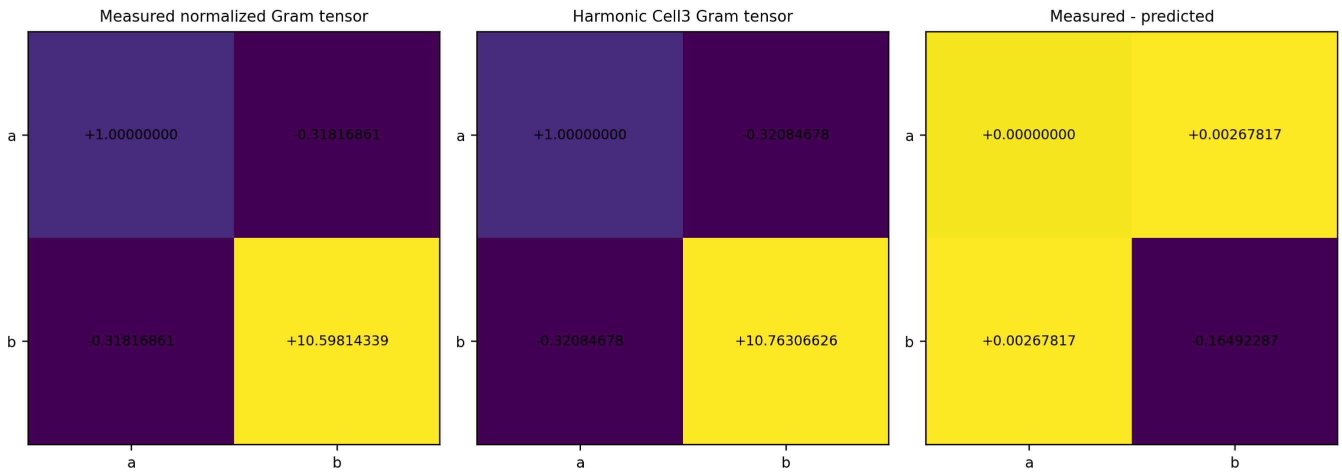


Figure 8. Normalized measured Gram tensor, harmonic Cell3 Gram tensor, and their residual. The angle enters through the off-diagonal cosine term; the dominant tensor difference comes from the b/a scale ratio, while the angular term is nearly locked.

This tensor view separates two geometric questions. The angle lock asks whether the normalized directions have the same internal separation. The edge-ratio comparison asks whether the anisotropic scale along those directions is the same. The data answer the first question much more tightly than the second.

# 7. Why the Angle Outranks Length and Area Comparisons

Absolute edge lengths depend on the selected recursive level r and harmonic orders m. Face area depends quadratically on those scales. The angle is unchanged by a common recursive enlargement and is much less sensitive to absolute scale assignment. This creates a natural hierarchy of evidence.

| Comparison | Residual | Dependence on recursive/harmonic scale | Interpretive rank |
|---|---|---|---|
| Included face angle | -0.003722778 deg = -13.402001 arcsec | Scale independent | Primary |
| a edge length | +0.135137% | Scale dependent | Secondary |
| b edge length | -0.635012% | Scale dependent | Secondary |
| edge ratio b/a | -0.769110% | Common-scale invariant but harmonic-assignment dependent | Secondary |
| face area | -0.500099% | Quadratically scale dependent | Secondary |

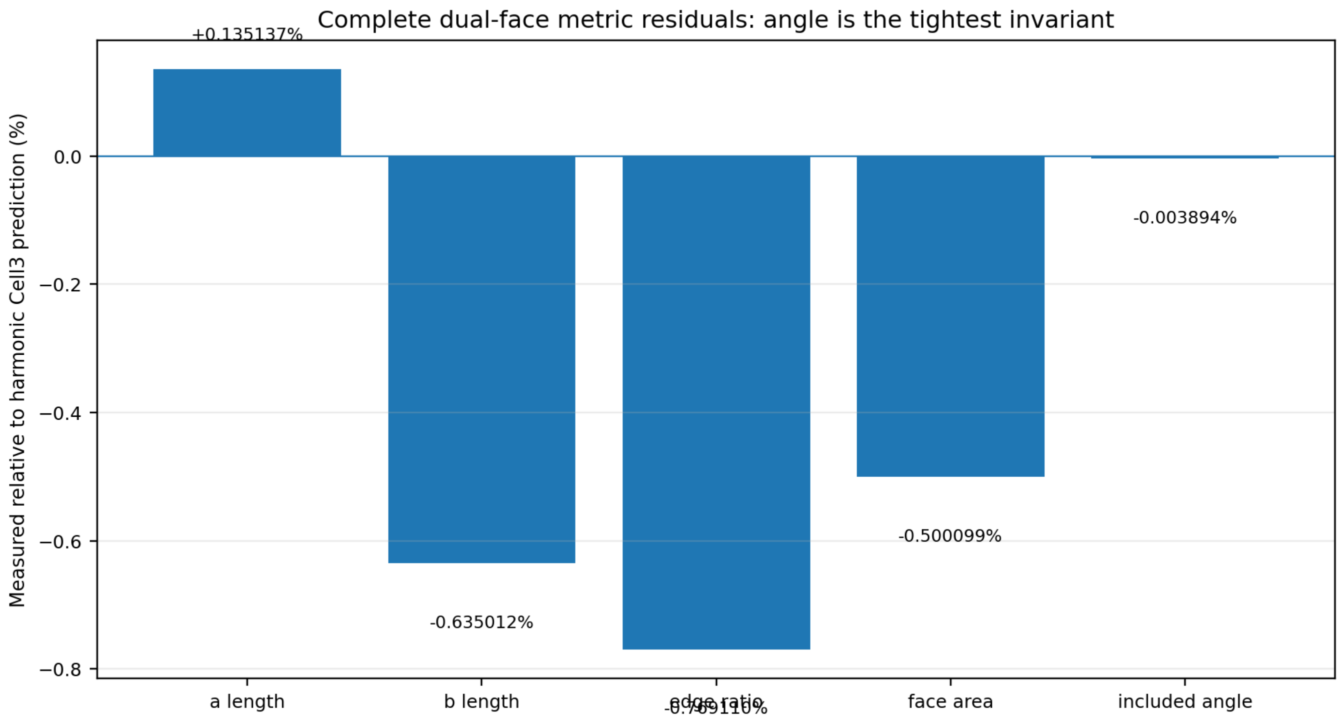


Figure 9. Complete dual-face metric residuals. The angle is shown as its percentage of the predicted angle so it can share the axis with length, ratio, and area residuals. Its relative magnitude is approximately -0.003894 percent, much smaller than the other face-metric residuals.

**Primary geometric statement. The near-lock establishes internal shape correspondence. The sub-percent length and area matches establish approximate scale correspondence. These are distinct results and should not be conflated.**

# 8. Common Zone and Three-Dimensional Orientation Context

The qA and qB planes intersect along one direct-space axis. Their measured cross product is 0.036385969 deg from the supplied CMB direction. Transforming that axis into authoritative Cell3 direct coordinates gives a ratio nearly identical to the CMB ratio. The nearby low-order integer direction is [7,2,-1].

```
Measured intersection Cell3 ratio = [1, 0.28506983, -0.13159608]
CMB Cell3 ratio = [1, 0.28464349, -0.13158840]
Low-order zone direction = [7, 2, -1]
```

A reciprocal plane (h,k,l) contains this zone when 7h + 2k - l = 0. The qA assignment (1,6,19) and the zone-preserving qB assignment (2,-9,-4) satisfy this law exactly.

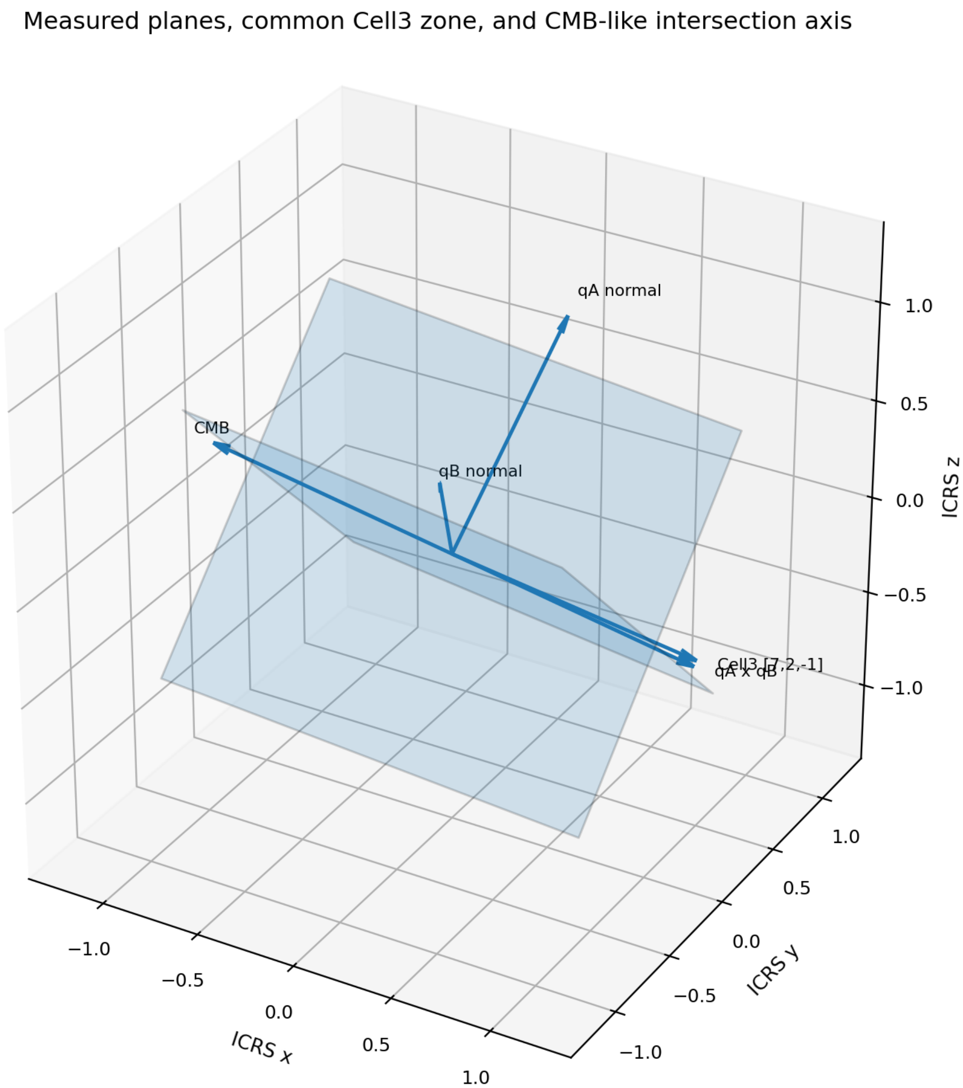


Figure 10. Three-dimensional orientation context. The qA and qB planes intersect along the measured longitudinal axis. The authoritative Cell3 [7,2,-1] direction and the supplied CMB axis nearly coincide with that intersection.

The zone relation supports the interpretation that the near-locked dual face is embedded in a common high-index Cell3 reciprocal zone. It does not determine the unresolved longitudinal period along the zone axis.

# 9. Supporting Harmonic Spacing Relationship

The qA and qF modes are close in direction and can be indexed to one high-index Cell3 family, (1,6,19). At recursive level r=18, qA occupies harmonic order 19 and qF occupies harmonic order 9. Their predicted spacing ratio is therefore 19/9.

```
dF/dA measured = 2.110603974823
19/9 target = 2.111111111111
relative residual = -0.024022245 percent
```

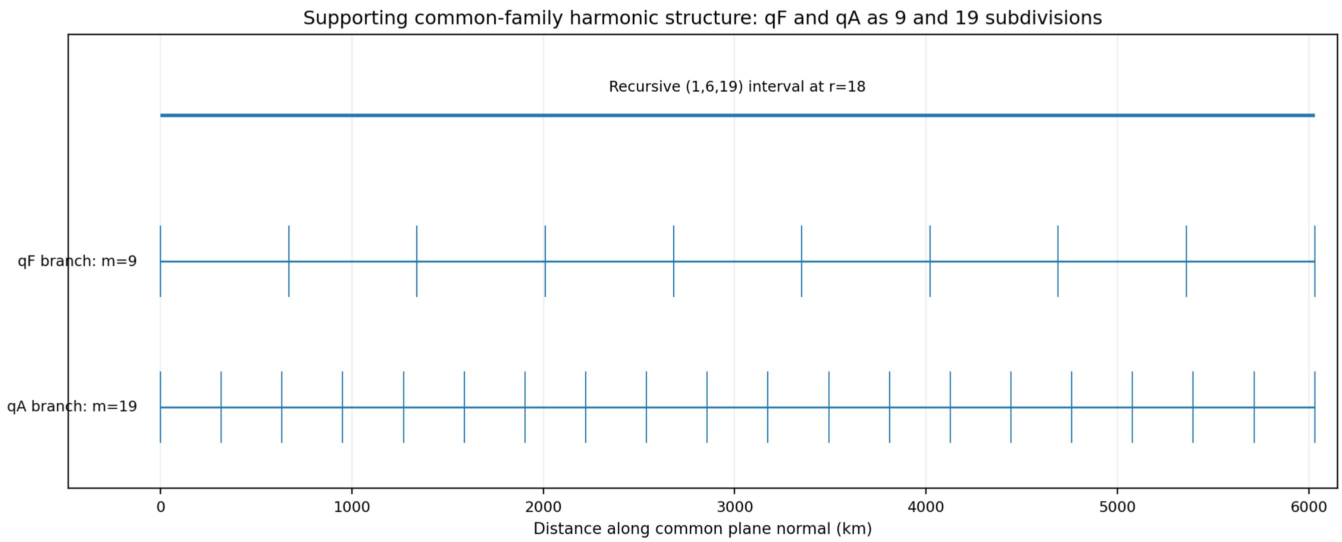


Figure 11. Exact recursive interval for the common Cell3 family (1,6,19) at r=18, divided into the qF m=9 and qA m=19 plane intervals. The observed spacing ratio differs from 19/9 by only -0.02402 percent.

This ratio is a strong supporting relation because it uses one shared indexed family. It remains secondary to the dual-face angle because the ratio requires the selected recursive and harmonic interpretation, whereas the measured face angle itself is obtained directly from the recovered reciprocal pair.

# 10. Supporting Reciprocal-Vector Sum Closure

The independently extracted broadband mode qC is close to qA + 2qB. The measured vector angle between qC and qA + 2qB is 0.73671 deg, and the spacing difference is approximately 0.20246 percent. A Cell3 integer-packet construction produces (17,-258,-433) at recursive level 18.

```
Cell3 packet qC predicted RA = 47.862525 deg
Cell3 packet qC predicted Dec = +68.908244 deg
orientation residual = 0.659366 deg
spacing residual = +0.116218 percent
```

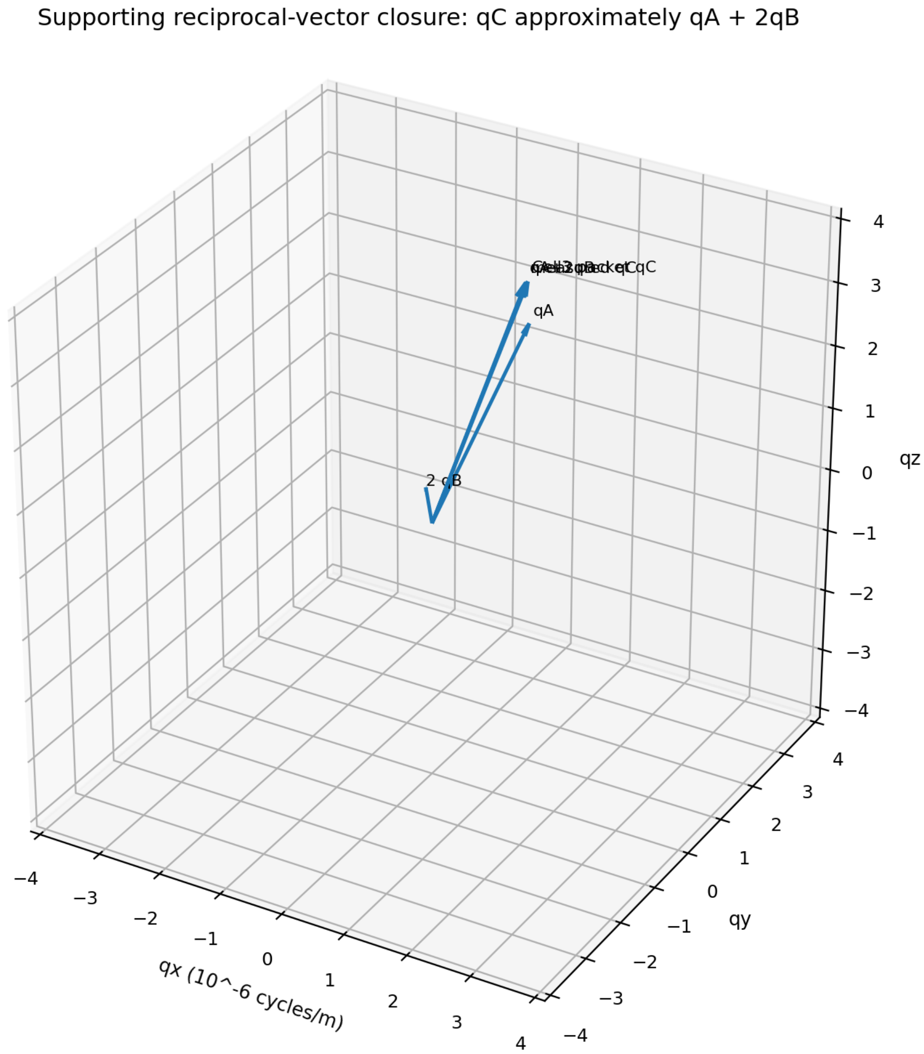


Figure 12. Reciprocal-vector closure. The independently measured qC vector, measured qA+2qB sum, and Cell3 integer packet prediction occupy the same local region in reciprocal space.

This closure shows that the dual-face plane pair participates in a larger reciprocal-vector grammar. It is supporting evidence for the same geometric system, not the basis of the angle calculation.

# 11. Educational Interpretation of the Near-Lock

## 11.1 What is directly measured

The spectrophotometer analysis supplies qA and qB as absolute ICRS reciprocal vectors. From those vectors, the dual direct edges and their included angle are calculated without choosing a Cell3 recursive level. The measured angle is therefore a property of the recovered signal geometry.

## 11.2 What is predicted from Cell3

The authoritative Cell3 basis supplies an exact reciprocal atlas. High-index Miller packets, recursive level, and harmonic order define a predicted qA/qB pair. Applying the same dual transformation to that predicted pair produces the harmonic Cell3 direct face and its angle.

## 11.3 What the near-lock means

The near-lock means that the relative geometry of the two measured plane families is almost the same as the relative geometry of the two harmonic Cell3 plane families. The result is not merely two similar distances. It is an agreement in the internal angle of a complete direct-space face recovered from reciprocal measurements.

## 11.4 What remains unresolved

Only the transverse face is recovered. The third direct period along the CMB-like zone axis is not measured. Therefore the complete three-dimensional macroscopic primitive volume, its volume, and its full set of three face ratios remain unresolved. The present result is an exposed two-dimensional face and zone.

**Foundational conclusion. The 13.402-arcsecond dual-face angular lock is the principal scale-independent bridge from the authoritative nanometre Cell3 basis to the measured macroscopic spectrophotometer plane structure.**

# 12. Reproduction Procedure

**1.** Load the authoritative Cell3 basis B and compute G = B^(-T).

**2.** Load measured reciprocal vectors qA and qB in ICRS cycles per metre.

**3.** Construct measured Q = [qA^T; qB^T] and D = Q^T(QQ^T)^(-1).

**4.** Extract measured direct edges a_exp and b_exp from the columns of D.

**5.** Compute the measured angle from the normalized dot product.

**6.** Generate harmonic Cell3 qA from (1,6,19), r=18, m=19.

**7.** Generate harmonic Cell3 qB from (1,-4,-2), r=18, m=18.

**8.** Apply the identical dual transformation to the harmonic pair.

**9.** Compute the predicted included angle and subtract it from the measured angle.

**10.** Convert the angular residual to arcseconds and radians.

**11.** Generate all figures and node ledgers from the same exact vectors.

**12.** Render the PDF with ReportLab core fonts and verify every page visually.

## 12.1 Exact calculation packet

```
theta_measured = 95.608657023061824 deg
theta_Cell3 = 95.612379800991320 deg
delta_theta = -0.003722777929497 deg
delta_theta = -13.402000546188 arcsec
delta_theta = -6.497473219029302e-05 rad
```

# Appendix A. Exact Direct-Face Vector Ledger

| Vector | x (m) | y (m) | z (m) | Length (m) | RA (deg) | Dec (deg) |
|---|---|---|---|---|---|---|
| measured a | 23580.891690751218 | 241209.358896143938 | 208024.244963551173 | 319393.330808073166 | 84.416444375 | +40.640484520 |
| measured b | 320888.448477832426 | -735846.439950803644 | 660832.415677104844 | 1039778.275001175469 | 293.561087947 | +39.460884024 |
| predicted a | 13074.285037135523 | 243490.051228412834 | 205617.616359058593 | 318962.295786866278 | 86.926434334 | +40.139103291 |
| predicted b | 369728.189380656870 | -716862.986621565535 | 666640.845742479432 | 1046423.190120965359 | 297.282806208 | +39.573399699 |

## Appendix A.1 Duality checks

| Check | Measured value |
|---|---|
| qA.a | 1.000000000000000e+00 cycles |
| qB.a | 3.084149929215548e-17 cycles |
| qA.b | -3.077339029617111e-17 cycles |
| qB.b | 9.999999999999999e-01 cycles |

# Appendix B. Complete 27-Node Authoritative Cell3 Ledger

| Address | Fractional coordinate | ICRS x (nm) | ICRS y (nm) | ICRS z (nm) | Radius (nm) |
|---|---|---|---|---|---|
| (-1, -1, -1) | (-0.5, -0.5, -0.5) | -0.176351350137 | +0.118682429905 | +0.637311498325 | 0.671826810838 |
| (-1, -1, 0) | (-0.5, -0.5, 0.0) | -0.181416008950 | -0.076211421278 | +0.292444294142 | 0.352482076442 |
| (-1, -1, 1) | (-0.5, -0.5, 0.5) | -0.186480667764 | -0.271105272461 | -0.052422910041 | 0.333198543969 |
| (-1, 0, -1) | (-0.5, 0.0, -0.5) | -0.158685962215 | +0.222342355125 | +0.363610955631 | 0.454785976632 |
| (-1, 0, 0) | (-0.5, 0.0, 0.0) | -0.163750621028 | +0.027448503942 | +0.018743751447 | 0.167089839530 |
| (-1, 0, 1) | (-0.5, 0.0, 0.5) | -0.168815279841 | -0.167445347241 | -0.326123452736 | 0.403600110809 |
| (-1, 1, -1) | (-0.5, 0.5, -0.5) | -0.141020574293 | +0.326002280345 | +0.089910412936 | 0.366398924014 |
| (-1, 1, 0) | (-0.5, 0.5, 0.0) | -0.146085233106 | +0.131108429162 | -0.254956791247 | 0.321765879067 |
| (-1, 1, 1) | (-0.5, 0.5, 0.5) | -0.151149891919 | -0.063785422021 | -0.599823995431 | 0.621855043707 |
| (0, -1, -1) | (0.0, -0.5, -0.5) | -0.012600729108 | +0.091233925963 | +0.618567746878 | 0.625386652479 |
| (0, -1, 0) | (0.0, -0.5, 0.0) | -0.017665387922 | -0.103659925220 | +0.273700542695 | 0.293205445206 |
| (0, -1, 1) | (0.0, -0.5, 0.5) | -0.022730046735 | -0.298553776403 | -0.071166661489 | 0.307759169054 |
| (0, 0, -1) | (0.0, 0.0, -0.5) | +0.005064658813 | +0.194893851183 | +0.344867204183 | 0.396159882521 |
| (0, 0, 0) | (0.0, 0.0, 0.0) | +0.000000000000 | +0.000000000000 | +0.000000000000 | 0.000000000000 |
| (0, 0, 1) | (0.0, 0.0, 0.5) | -0.005064658813 | -0.194893851183 | -0.344867204183 | 0.396159882521 |
| (0, 1, -1) | (0.0, 0.5, -0.5) | +0.022730046735 | +0.298553776403 | +0.071166661489 | 0.307759169054 |
| (0, 1, 0) | (0.0, 0.5, 0.0) | +0.017665387922 | +0.103659925220 | -0.273700542695 | 0.293205445206 |
| (0, 1, 1) | (0.0, 0.5, 0.5) | +0.012600729108 | -0.091233925963 | -0.618567746878 | 0.625386652479 |
| (1, -1, -1) | (0.5, -0.5, -0.5) | +0.151149891919 | +0.063785422021 | +0.599823995431 | 0.621855043707 |
| (1, -1, 0) | (0.5, -0.5, 0.0) | +0.146085233106 | -0.131108429162 | +0.254956791247 | 0.321765879067 |
| (1, -1, 1) | (0.5, -0.5, 0.5) | +0.141020574293 | -0.326002280345 | -0.089910412936 | 0.366398924014 |
| (1, 0, -1) | (0.5, 0.0, -0.5) | +0.168815279841 | +0.167445347241 | +0.326123452736 | 0.403600110809 |
| (1, 0, 0) | (0.5, 0.0, 0.0) | +0.163750621028 | -0.027448503942 | -0.018743751447 | 0.167089839530 |
| (1, 0, 1) | (0.5, 0.0, 0.5) | +0.158685962215 | -0.222342355125 | -0.363610955631 | 0.454785976632 |
| (1, 1, -1) | (0.5, 0.5, -0.5) | +0.186480667764 | +0.271105272461 | +0.052422910041 | 0.333198543969 |
| (1, 1, 0) | (0.5, 0.5, 0.0) | +0.181416008950 | +0.076211421278 | -0.292444294142 | 0.352482076442 |
| (1, 1, 1) | (0.5, 0.5, 0.5) | +0.176351350137 | -0.118682429905 | -0.637311498325 | 0.671826810838 |

# Appendix C. Complete Exposed Dual-Face Node Ledger

| Address | Measured x (m) | Measured y (m) | Measured z (m) | qA phase | qB phase | Predicted radius (m) |
|---|---|---|---|---|---|---|
| (-1, -1) | -344469.340169 | +494637.081055 | -868856.660641 | -1.000000000000 | -1.000000000000 | 1063698.498208 |
| (-1, 0) | -23580.891691 | -241209.358896 | -208024.244964 | -1.000000000000 | -0.000000000000 | 318962.295787 |
| (-1, 1) | +297307.556787 | -977055.798847 | +452808.170714 | -1.000000000000 | 1.000000000000 | 1123397.695753 |
| (-1, 2) | +618196.005265 | -1712902.238798 | +1113640.586391 | -1.000000000000 | 2.000000000000 | 2147629.112570 |
| (0, -1) | -320888.448478 | +735846.439951 | -660832.415677 | 0.000000000000 | -1.000000000000 | 1046423.190121 |
| (0, 0) | +0.000000 | +0.000000 | +0.000000 | 0.000000000000 | 0.000000000000 | 0.000000 |
| (0, 1) | +320888.448478 | -735846.439951 | +660832.415677 | -0.000000000000 | 1.000000000000 | 1046423.190121 |
| (0, 2) | +641776.896956 | -1471692.879902 | +1321664.831354 | -0.000000000000 | 2.000000000000 | 2092846.380242 |
| (1, -1) | -297307.556787 | +977055.798847 | -452808.170714 | 1.000000000000 | -1.000000000000 | 1123397.695753 |
| (1, 0) | +23580.891691 | +241209.358896 | +208024.244964 | 1.000000000000 | 0.000000000000 | 318962.295787 |
| (1, 1) | +344469.340169 | -494637.081055 | +868856.660641 | 1.000000000000 | 1.000000000000 | 1063698.498208 |
| (1, 2) | +665357.788646 | -1230483.521005 | +1529689.076318 | 1.000000000000 | 2.000000000000 | 2085947.034249 |
| (2, -1) | -273726.665096 | +1218265.157743 | -244783.925750 | 2.000000000000 | -1.000000000000 | 1277699.951119 |
| (2, 0) | +47161.783382 | +482418.717792 | +416048.489927 | 2.000000000000 | 0.000000000000 | 637924.591574 |
| (2, 1) | +368050.231859 | -253427.722159 | +1076880.905604 | 2.000000000000 | 1.000000000000 | 1171059.942798 |
| (2, 2) | +688938.680337 | -989274.162109 | +1737713.321281 | 2.000000000000 | 2.000000000000 | 2127396.996417 |

# Appendix D. Archival and Reproduction Record

| Field | Value |
|---|---|
| Principal Investigator | P. Dwayne Esterline |
| Degrees | BS, Huntington University (1999); MBA, Indiana Wesleyan University (2008) |
| Organization | Krytronx, LLC. |
| Classification | Proprietary Confidential Information and Trade Secret of Krytronx, LLC. All Rights Reserved. Do Not Distribute. |
| Generation epoch | 1785939329 |
| Generation UTC | 2026-08-05T14:15:29Z |
| PDF | K-Field_cell3_to_spectrophotometer_near_lock_face_angle_1785939329.pdf |
| JSON | K-Field_cell3_to_spectrophotometer_near_lock_face_angle_1785939329.json |
| PDF engine | Python ReportLab |
| PDF fonts | Helvetica, Helvetica-Bold, Times-Roman, Courier only |
| Historical format references | K-Field Cell3 to Spectrophotometer Planar Comparison; K-Field Spectrophotometer Optimized Geometry; K-Field Spectrophotometer Hidden Structure Energy Analysis |
| Machine-readable scope | Exact angle-lock metrics, dual vectors, normalized Gram tensors, Cell3 basis and 27 nodes, measured planes, harmonic assignments, zone geometry, supporting closures, source hashes, and figure manifest |

## Appendix D.1 Source hash ledger

| Source file | SHA-256 | Bytes |
|---|---|---|
| K-Field_cell3_to_spectrophotometer_planar_comparison_1785937320.json | 3100ea7414fdf104697df366cd299e0f069a7ba8b32ef8f7b917fabbd1f5573b | 48784 |
| K-Field_spectrophtometer_pic_id14E_analysis_exposed_plane_structure_1785932820.json | 79e3aa2b187a9ad4fbd2b481de1b22ff7c53753c738678f3f593d3c0fc8e3c85 | 1545939 |
| K-Field_ICRS_plane_scew_rajectory_analysis_1785606737.json | 03e6d901dbbe0801cada29a559d40b50c3cec5360f8e29f96d80e8e818cae3c3 | 47024 |
| pic_id14E_full24_hrs_20260805.csv | 6f5e3c69c623fe02b854e121110f915861bab24e8b6f000b094e810f49b53f62 | 31949154 |
| K-Field_cell3_to_spectrophotometer_planar_comparison_1785937320.pdf | ec31b4106be203c4836fab3d322a345e54f80f07b7f2feb549bacb5a2a306337 | 2380860 |

**END OF PROPRIETARY CONFIDENTIAL TRADE SECRET ARCHIVAL REFERENCE**